

Ian Leung

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EDUCATION

University of Waterloo

Sep 2025 – Present

Bachelor of Computer Science and Finance - Double Major

- President's Scholarship of Distinction (**\$5000**), Undergraduate Achievement Award (**\$1500**)
- Statistics Minor

Relevant Coursework: Functional Programming (Advanced), Interpreters (Advanced), Linear Algebra, Calculus, Probability, Proofs, Financial Markets and Data Analytics

TECHNICAL SKILLS

Languages Python, Java, C++, C#, C, Go, Kotlin, TypeScript, SQL, HTML, CSS, Bash
Technologies Temporal, PyTorch, Supabase, Firebase, MongoDB, React, Maven, Qt, OpenRouter, OpenCV
Developer Tools Linux, Git, Docker, Jupyter, DigitalOcean, Modal, Render, Vercel, Unity, Android Studio

WORK EXPERIENCE

Roblox

San Mateo, CA

Software Engineer Intern | Go, Python

Jun 2026 – Present

- Building an automated CI/CD pipeline that deploys configuration changes across **5,000+** global network devices (servers) using **GitHub Actions** and **Temporal**, replacing a slow, manual, error-prone process
- Designing a **canary-first** rollout with automated health checks (**Pingmesh**, **VictoriaMetrics**) and staged, gated expansion, validating changes on a small device set before fleet-wide deployment to Roblox's **100M+** daily active users
- Implementing deployment safety controls including device **locking**, automatic **rollback**, and error classification, so devices self-recover to a known-good state on failure

Sunnybrook Research Institute

Toronto, ON

Software Research Intern | Python

Jul 2025 – Aug 2025

- Built an event tracing and playback tool to simulate medical professionals' interactions with **Focused Ultrasound** 🏠 software, enabling realistic testing and diagnosis of software defects, unintended user behaviours, and edge cases
- Created an automated software testing framework with **PyQt** and **Pydantic**, generating synthetic and replayable JSON event flows, reducing software testing times by **91%**
- Implemented a logging API using **Difflib** and **Loguru** to validate test results against baselines, identifying **5+** bugs

PROJECTS

🌐 Data Science Club Website | TypeScript, React, Supabase, HTML, CSS

- Maintained www.uwdatascience.ca, UWaterloo's data science club website serving **600+** members
- Replaced the executive team's manual term-start **Google Form** with an in-app onboarding flow backed by a normalized **Supabase/Postgres** schema that autofills profile and role data from existing records

🏠 Pharmagotchi: Pharmacy Companion App | Android Studio, Figma, Kotlin, XML, OpenRouter

- Built a native **Android** (Kotlin) health-companion app where a virtual pet responds to missed medication doses, concerning vitals, and skipped logging, backed by an offline **Room** database and **WorkManager** background reminders
- Integrated **LLMs via OpenRouter** to power a context-aware chat assistant and auto-build each user's tracked metrics, parsing their conditions and medications into structured **JSON** vital signs

🏠 PeerAssist - Full-Stack Peer Reviewing | Java, MongoDB, Apache PDFBox, Maven

- Created a full-stack **Java** desktop application for 20+ students to upload, review, and grade peers' academic work
- Integrated a **Supabase** backend for authentication and data storage, enforcing **Row-Level Security** policies across profiles, documents, and reviews tables to control per-user access
- Rendered uploaded PDFs with **Apache PDFBox** and added search, sorting, and ranking by average review score; packaged the app into a single zero-config executable JAR via **Maven** shade plugin

🏠 Trust No Ghost - 3D Maze Exploration Game | Unity, C#, Itch.io

- Shipped Trust No Ghost 🏠 for **60+** users at the UW Game Jam Fall 2025, placing in the **top 10** most rated games
- Harnessed Unity and C# to develop expanding maze designs, AI pathfinding algorithms, and dynamic moving mechanics